

Mehul Kaushik

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LinkedIn — LeetCode — GitHub — Portfolio

Education

- **Manipal Institute of Technology (MIT)** **Udupi, India**
• **Bachelor of Technology, Computer Science** **2023 – 2027**
 - **Relevant Coursework:** Machine Learning, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence

Summary

Aspiring Software Development Engineer seeking to apply expertise in programming, data structures, and full-stack and AI-driven application development to drive innovation and deliver impactful solutions through effective communication.

Experience

- **ISTE Manipal** **Manipal, India**
• **Head of Cybersecurity** **July 2025 – July 2026**
 - Coordinated **CTF competitions** for over **150 participants**, crafting challenges on web security.
 - Directed a **10-member team** conducting workshops and mentorship sessions.
- **STMicroelectronics** **India**
• **Python Automation Intern** **May 2025 – June 2025**
 - Automated reporting workflows using **Python**, reducing manual effort by **60%**.
 - Developed modular **Python** automation scripts for the **CVIP PLL** verification layer, reducing manual verification effort by **40%** and accelerating regression workflows by 2 times.

Projects

- **AskPDF – RAG-Powered Document Q&A System** [GitHub] **Node.js, MongoDB, Redis, Gemini API**
 - Built a **RAG system** that enables users to upload PDFs and query them via a conversational LLM interface – answers are grounded strictly in document content with **real-time streaming**.
 - Processed documents through a **5-stage async pipeline** (parse → chunk → embed → index → search), efficiently handling PDFs up to **20MB** and generating **768-dimensional embeddings** for semantic vector search.
 - Offloaded PDF processing to a **decoupled Redis worker** using a **BRPOP** blocking queue, enabling the API to return **202 Accepted** immediately while tracking asynchronous job status via **Redis Hashes (HSET)**.
 - Engineered **hallucination mitigation** via similarity thresholding (≥ 0.7 cutoff) to reject low-relevance context before LLM generation, and implemented automated state cleanup to prevent orphan records in MongoDB during async worker failures.
- **MoonMovement – Startup Community Platform** [Live] [GitHub] **PostgreSQL, Prisma, Redis, AWS, Nginx**
 - Built a **Reddit-style startup community platform** supporting startup discovery, threaded discussions, voting, user profiles, startup directories, and role-based moderation through a scalable REST API architecture.
 - Designed a **3-tier backend architecture** with stateless API servers, PostgreSQL, and Redis caching, deploying the application on **AWS EC2** with **Nginx** reverse proxy and **AWS RDS** for managed database hosting.
 - Engineered low-latency content delivery using **Redis** for session management, hot-feed caching, and rate limiting while optimizing complex Prisma queries to eliminate N+1 lookups and reduce database load.
 - Implemented secure authentication, RBAC, and asynchronous notification workflows, enabling scalable concurrent request handling with efficient database indexing, pagination, and transactional consistency.

Technical Skills

- **Programming Languages:** C++, Python, Java, SQL
- **Backend:** Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, Prisma, Redis
- **Cloud & DevOps:** AWS (EC2, RDS), Docker, Nginx, Git, Linux
- **Machine Learning & AI:** RAG, LLM Applications, NLP, Vector Search

Achievements

- **Competitive Programming:** Solved **500+** LeetCode problems, achieved the **Knight** badge.
- **Amazon ML Challenge:** Secured **AIR 318** among participants nationwide in the Amazon Machine Learning Challenge.
- **Hackathons:** Won **3 Hackathons** and was selected as a **Smart India Hackathon (SIH) Finalist**.
- **Open Source:** Contributed to the **HumanAI** project as part of **Google Summer of Code**, with **1** pull request merged.

Certifications

- **Andrew Ng's Machine Learning Specialization (Coursera):** Covered supervised and unsupervised learning, neural networks, model evaluation, and end-to-end machine learning workflows using Python.